MATHANKUMAR J

AI Engineer · Coimbatore, Tamil Nadu, India
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SUMMARY

AI Engineer building and deploying production LLM and agentic systems inside client environments. Retrieval-augmented generation end to end, retrieval evaluation scored against real corpora, and agent workflows with the constraints needed to run them unattended. Google Cloud Professional Cloud Developer and Claude Certified Architect.

TECHNICAL SKILLS

Languages and frameworks: Python, FastAPI, SQLAlchemy, Django, JavaScript, React
LLM and agents: RAG, embeddings, retrieval evaluation, constrained decoding (GBNF), prompt engineering, LangChain, LangGraph, LlamaIndex, MCP and tool calling, local inference (llama.cpp, ONNX), OpenAI and Anthropic APIs
Data: PostgreSQL, Redis, Qdrant, Neo4j, MongoDB
Platform: Docker, Azure, GCP, CI/CD, observability, WebSocket and WebRTC

EXPERIENCE

AI Engineer | Techjays
Jul 2026 – Present · Coimbatore, India

- Designed a multi-process AI system behind a single audited gateway, with model routing held as configuration rather than hard-coded constants and prompts versioned as files.
- Built retrieval evaluation into the RAG pipeline rather than assuming it: ranking scored against a real corpus, broken down per bucket, with the limits of the measurement documented alongside the results.
- Designed agent workflows in LangChain and LangGraph with MCP tool integrations and human-in-the-loop checkpoints on any action that carries cost or leaves the organisation.
- Built the safeguards around agent-written code: enforcement hooks, a pull request gate with its own test suite, read-only review agents running in parallel, and a prompt-injection corpus in CI.
- Cut database load through set-based query design against a measured load model.

Software Engineering Analyst | Techjays
Dec 2025 – Jul 2026 · Coimbatore, India

- Built AI-powered applications in Python and Django covering voice AI and text generation, integrating OpenAI APIs.
- Implemented real-time communication systems using WebSocket and WebRTC.

Full Stack Developer (Internship) | Techjays
Jul 2025 – Dec 2025 · Remote

- Built features across MERN-stack projects in an agile team.

Full Stack Developer | Freelance
Sep 2024 – Jul 2025 · Coimbatore, India

- Delivered web applications across frontend, backend, API integration and database design.

PROJECTS

clause — github.com/Mathan-j/clause
Regulatory answer engine over 61 RBI circulars where every factual sentence resolves to an exact character span in an exact document. A GBNF grammar constrains decoding so the model can only cite passages by index; citation text is sliced from the corpus, and an answer whose citation fails to resolve is discarded. recall@5 0.446 against a chance floor of 0.084, 8.3% of drafts rejected by the citation contract, support signal 0.822. Refusal gate caught 0 of 15 unanswerable questions — published as a measured limitation. Python, FastAPI, PostgreSQL, Qdrant, llama.cpp, Docker.

triage-lab — github.com/Mathan-j/triage-lab
Support-ticket triage pipeline built stage by stage: ingest, WordPiece chunking, MiniLM embeddings, retrieval. recall@5 0.76, MRR 0.5190, a lift of 0.1759 over baseline ranking, 261 tests. Python.

embed-lab — github.com/Mathan-j/embed-lab
Inspection tool for what a language model does with a word: tokens, 384-dimension embedding and nearest neighbours, served through ONNX runtime inference. 37 tests. Python.

EDUCATION

Master of Computer Applications, Computer Science
Bharathiar University — Bishop Appasamy College of Education, Coimbatore · 2023 – 2025

CERTIFICATIONS

Google Cloud Certified — Professional Cloud Developer, Google Cloud · Feb 2026 – Feb 2028
Claude Certified Architect — Foundations, Anthropic · Aug 2026 – Aug 2027